

QM35 DK-05 Release Notes

Qorvo

Release 3.0.0

Introduction

Overview of the Release

The QM35 SDK release marks the initial availability of Qorvo®'s UWB development toolkit for early customer evaluation. This release provides the foundational components necessary for working with the QM35 chipset, including core software components, API, and initial documentation. The release focuses on enabling essential features such as ranging and sensing aimed at consumer applications of UWB technology.

It includes the basic infrastructure for UWB operations and introduces the initial version of the key API called Cherry based on the FiRa UCI protocol.

It primarily targets initial trials, proof-of-concept implementations, and internal evaluations, with ongoing enhancements planned for future versions.

Purpose of the SDK

The purpose of the QM35 SDK is to provide a scalable, modular, and user-friendly software development platform that accelerates the adoption of Qorvo®'s UWB solutions across multiple market segments. The QM35 SDK is designed to facilitate the integration of UWB technology into a wide range of applications by offering key building blocks, reference designs, and tools. It simplifies the development process for customers by delivering a comprehensive set of APIs, sample code, and documentation tailored to specific use cases such as, but not limited to:

Table 1.1: Use-cases

Use-case	UWB feature	Status
Precise distance measurement	Two-way ranging (TWR)	Supported
Locating device in 2D angular position	2D AoA (Angle of Arrival)	Supported
Human presence detection	UWB Sensing	Supported
Indoor Navigation	DL-TDoA	Supported

This QM35 SDK is intended to support the evaluation, prototyping, and initial product development stages, enabling customers to quickly validate their ideas and bring UWB-enabled products to market. By leveraging the QM35 SDK, users can focus on application-specific challenges while relying on Qorvo®'s robust and optimized UWB stack to handle the complexities of wireless communication and localization. This release aims to support various customer personas, from engineers experimenting with early-stage prototypes to developers integrating UWB features into production-ready systems.

SDK Components

- QM35 Firmware: Precompiled firmware binary tailored for the QM35825 chip, essential for operating and evaluating the chip's core functionalities.
- Cherry: A C-based API that delivers direct access to the QM35 chip's features, providing developers with the necessary libraries for building custom UWB applications.
- UWB Qorvo Tools: A Python-based framework primarily intended for evaluation purposes, featuring sample code that helps users quickly assess the basic functionalities of the QM35 chip.
- Linux-drivers: Source code for Linux drivers that enable smooth communication between the host system and the QM35 chip, essential for UWB application development on Linux platforms.
- Embedded Zephyr: Platform-specific nRF drivers sources. These can be used to develop custom applications for the QM35825 chip on nRF platform.
- QSoC-Flash: A cross-platform tool supporting both Windows and Linux systems, designed to simplify the process of flashing the QM35 chip via the SPI interface, making firmware updates more accessible and manageable.
- RPi Image: Qorvo® Raspberry Pi OS image, providing a ready-to-use system to evaluate and develop for QM35825 using RPi 4 host.
- UWB Explorer App: A cross-platform GUI application for evaluating QM35 UWB features, compatible with both Windows and macOS.
- Documentation: Initial documentation to onboard customers to evaluate QM35-based design kits.

Target Audience

- Embedded Software Developers working on UWB applications
- System Architects designing UWB-based solutions
- Field Application Engineers (FAEs) supporting UWB integration
- Product Managers evaluating UWB for market opportunities
- R&D Teams developing proof-of-concept (POC) solutions
- OEMs and ODMs focused on IoT, smart home, and industrial automation
- Early-stage customers exploring UWB technology for ranging and Radar use cases
- Technology Partners and Solution Providers integrating UWB into broader ecosystems

Release Version and Date

Table 2.1: Release version

SDK Version Number	Release Date
v3.0.0	2025-07-17

DK Components Revisions

SDK

Table 3.1: SDK Components Revisions

Component	Revision
QM35 Firmware	R13.1.0
Cherry	R13.1.0
UWB Qorvo Tools	R13.1.0
Linux-drivers	R13.1.0
Embedded Zephyr	R13.1.0
QSoC-Flash	v4.1.0
Explorer App	v1.0.0
RPi Image	R0.2.5

HDK

Table 3.2: HDK Components

Component	Revision
QM35825.E1.x DK-05 (2 x QM35825 + PC Interface Board + RPi Interface Board + RPi Board)	Rev E

Features Supported

		Explorer App	UQT (Python)		Cherry		FW/UCI
			Sample App	API	Sample App	API	
FiRa	TWR	✓	✓	✓	✓	✓	✓
	DL-TDOA	✗	✓	✓	✓	✓	✓
	OWR AM	✗	✓	✓	✗	✗	✓
	Data transfer	✗	✗	✗	✗	✗	✓
	HUS	✗	✗	✗	✗	✗	✓
Radar CIR		✓	✓	✓	✓	✓	✓
Interleaving Radar CIR and TWR		✗	✓	✓	✗	✓	✓
Radar presence detection		✓	✓	✓	✗	✓	✓

Figure 4.1: Supported features matrix

Key Features

QM35 firmware

- **Fira Protocol Support**
 - **FiRa TWR** (One to One, One to many): This is a method where two devices exchange timestamped messages to accurately calculate the distance between them based on the round-trip time of the signals.
 - **FiRa DL-TDOA**: This is a positioning method that enables DT-Tags to estimate their own locations based on the reception of DTMs from DT-Anchors.
 - **FiRa OWR**: This enables distance and angle measurements between devices using unidirectional broadcast messages, eliminating the need for two-way communication channels.
 - **Hybrid Scheduling (HUS)**: This is a feature that allows scheduling UWB sessions configured in Time-scheduled mode or Contention-based mode with a fixed time relation to each other such that they occur in the HUS Ranging round in a deterministic time sequence.
 - **Data Transfer**: This mode enables devices to exchange information by embedding data within UWB frames, allowing for secure, low-power, and high-speed short-range communication.
- **2D 360 AoA:**
 - **360° AoA measurement**: This is a specific mode which allows measuring a 360° AoA. It is based on the three pairs made by three antennas arranged in a triangle. It is also known as PDOA Mode 1. This functionality requires **Torus antennas** which are **not included** in the QM35825DK-05 kit. To obtain them, contact [Qorvo sales](https://www.qorvo.com/support/how-to-buy/contact-a-sales-rep)¹.
- **Low Power Mode:**
 - **Power Management**: The firmware automatically handles its power management.
- **Radar:**
 - **Simultaneous transmission**: The Radar can simultaneously transmit and receive frames.
 - **CIR extraction**: The Radar CIR is extracted with the lowest noise output.
- **Interleaving sessions:**
 - **Concurrent sessions**: The Radar and FiRa TWR can be run concurrently.

¹ <https://www.qorvo.com/support/how-to-buy/contact-a-sales-rep>

Cherry

- **Supported protocols:** FiRa TWR, FiRa DL-TDOA, and Radar.
- **Secure and flexible resource management:** Dedicated APIs for session creation and destruction.
- **Robust API:** With error management, multi-threading, and race condition handling.
- **Comprehensive examples:** Demonstrating the usage of the Cherry features and APIs and providing practical guidance for developers implementing UWB applications using Cherry.
- **Adaptable APIs:** With configurable parameters for ranging and session management.

UWB Qorvo Tools

- **Supported protocols:** FiRa TWR, FiRa DL-TDOA, OWR AoA Measurement, and Radar.
- **radar presence detection demo:** With up-to-date antenna management.
- **Interleaving sample:** For Radar and FiRa TWR concurrent sessions.
- **Antenna Calibration:** Calibration scripts are available for QM35825 antenna calibration.
- **360 2D AoA evaluation:** A simple script to evaluate the 360 2D AoA. This functionality requires **Torus antennas** which are **not included** in the QM35825DK-05 kit. To obtain them, contact [Qorvo sales](#)².

UWB Explorer App

- **Supported protocols:** FiRa TWR, Radar CIR and Radar presence detection
- **Ranging:** A GUI for ranging settings and results display

² <https://www.qorvo.com/support/how-to-buy/contact-a-sales-rep>

Known limitations

QM35 firmware

- Low Power Mode is automatically disabled when Sensing/Radar is active, as enabling it would degrade CIR (Channel Impulse Response) quality. The root cause has been identified, and this limitation is expected to be resolved in upcoming releases.

Cherry

- N/A

UWB Qorvo Tools

- UWB Qorvo Tools Guide uses the old MAC address format (mac addresses as strings, eg. "[00:a1", "00:a2"]") and destination MAC address format (mac addresses as complex nested tuples with round and slot information [(round_idx, [(mac, slot), ...]), ...]).
- When using the PC Interface Board, the QM35825 SoC takes more than 65ms to enter S3 power state after a ranging round. As a result, it is not recommended to use this platform for meaningful low power measurement.

UWB Explorer App

- The DK-06 is not supported.
- Localization app - FiRa Configuration:
 - Contention based schedule mode is not supported.
 - PSDU data rate is limited to 6.81 Mbps.
 - Number of STS segments is restricted to 1.

Linux Drivers

- The onboard power regulators are currently unmanaged by software and will remain turned on for the QM35825 operation.

RPi Image

- Due to statically assigned IP address and machine name, the SSH connection fails when multiple QM35825 + RPi running with Debian image are connected. To allow the connection of several RPi on the same machine, name and IP address of the RPi must be updated.

Compatibility

Supported Hardware Platforms

Table 7.1: Supported Hardware Platforms

Platform	Revision
QM35825.E1.x DK-05 (2 x QM35825 + PC Interface Board + RPi Interface Board + RPi Board)	Rev E

Supported Operating Systems

Table 7.2: Supported Operating Systems

System	Version
Ubuntu	22.04.4 LTS
Windows	10/11

Documentation

Table 8.1: New Documentation

Document	Version	Description	Update
Quick Start Guide	3.0.0	Guide to get started quickly with the QM35 SDK.	Document added.
QM35 High-Speed SPI Protocol	3.0.0	Detailed protocol documentation for high-speed SPI communication with the QM35 chip.	Document added.
QM35 Linux Kernel Driver Guide	R3.8.0	Instructions and guidance for using the Linux kernel drivers with the QM35 chip.	Document added.
UWB Qorvo Tools guide	R13.1.0	Detailed guide for using the UWB Qorvo Tools Python framework.	Document added.
UWB FiRa Protocol	R12.34.0	Protocol documentation for implementing FiRa functionalities with the QM35 chip.	Document added.
UWB Command Interface Specification	R12.34.0	Specifications for the UWB command interface used with the QM35 chip.	Document added.
UWB L1 configuration	R12.34.0	Configuration guide for Layer 1 (L1) UWB functionalities.	Document added.
UWB Radar Protocol	3.0.0	Protocol documentation for implementing Radar functionalities with the QM35 chip.	Document added.
QM35825DK-05 HW User Guide	v3.0.0	Hardware user guide for the PC setup and RPi development kits.	Document added.
Qorvo Software License Agreement	v3.0.0	Qorvo® Software License Agreement for the SDK v3.0.0.	Document added.
Cherry API and Integration Guide	R13.1.0	Detailed guide for using the Cherry API to implement UWB use-cases.	Document added.

Tools and Utilities

Table 9.1: New Tools

Tool	Revision	Description
QSoC-Flash	v4.1.0	Multiplatform tool that utilizes an SPI connection and the HSSPI protocol to flash firmware onto the QM35 chip.
Pi4-Setup	v1.0.0	Pi4-Setup is a setup tool to smoothly enable Raspberry Pi 4 QM35 SDK evaluation.

Regulatory and Compliance Updates

Compliance with standards

Table 10.1: Compliance

Standard	Version	Status
FiRa	v3.0	Compliant
FiRa	v2.0	Compliant
FiRa	v1.3	Deprecated

Support and Contact Information

Getting support and contacting Qorvo®

For the latest specifications, additional product information, worldwide sales and distribution locations:

Web: www.qorvo.com

Tel: 1-844-890-8163

Email: customer.support@qorvo.com

For additional technical assistance, visit and participate in our community forum at <https://forum.qorvo.com/>.

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Appendices

Glossary

AoA

Angle of Arrival - A technique used to determine the direction of arrival of a signal, particularly useful in UWB applications for determining the angular position of devices.

API

Application Programming Interface - A set of functions and procedures that allow the creation of applications which access the features or data of an operating system, application, or other service.

CIR

Channel Impulse Response - A representation of the signal's behavior as it travels through the communication channel, used in UWB for ranging and sensing applications.

DK

Development Kit - A collection of hardware tools provided by a manufacturer or vendor to help developers create applications or firmware for a specific platform, chip, or product family.

DL-TDoA

Downlink Time Difference of Arrival - A positioning technique used in wireless systems where a device calculates its position by measuring the time differences at which downlink signals arrive at the receiver.

FTDI

Future Technology Devices International - A company that designs and supplies USB and serial communication devices, often used in embedded systems for programming and debugging.

HDMI

High-Definition Multimedia Interface - A digital interface used to transmit audio and video data from a source device to a display or other output device.

HSSPI

High-Speed Serial Peripheral Interface - A high-speed communication protocol used to connect microcontrollers with peripherals, providing faster data transfer rates than standard SPI.

MAC

Medium Access Control - A sublayer of the data link layer that controls how devices in a network gain access to a medium and permission to transmit data.

UCI

UWB Subsystem Command Interface - The interface used to communicate with the UWB subsystem, providing commands and responses for UWB operations.

UWB

Ultra-Wide Band - A radio technology that uses a very low energy level for short-range, high-bandwidth communications over a large portion of the radio spectrum.

RAM

Random Access Memory - Volatile memory used for temporary data storage while a device is running. It is erased when device is powered off.

RSSI

Received Signal Strength Indicator - A measurement of the power present in a received radio signal, used to estimate the distance between devices.

SPI

Serial Peripheral Interface - A synchronous serial communication protocol used to connect microcontrollers with peripherals like sensors and memory.

SSH

Secure Shell - A cryptographic network protocol used for secure data communication, remote command-line login, and other secure network services over an unsecured network.

TWR

Two Way Ranging - A ranging technique that measures the round-trip time of signals between two devices to calculate the distance between them.